



REV. J · FORMULATION PACK

Open-Cell and Closed-Cell Foam

Formulas first. **BX-01** is the BazzLab card — the mix others do not publish. OC-01 and CC-01 stay as lab screens. TDS, catalyst, memo and patent notes follow so every number can be checked.

01 Best formula · BX-01 / BX-02

HFO-1233zd closed-cell on TB-307 / TEROL 250. Costs less than CREAPOL 326 / 1500. No Mannich. Unique to this floor.

01a 50 g lab cup + checklist

50 g CC (BX-01 + HFO) and 50 g OC (BX-02, no HFO), each with a tick-off bench list.

02 Open-cell formula · OC-01 / OC-02

High-water memo screen. Soft, air-open, 7–12 kg/m³.

03 Closed-cell formula · CC-01 / CC-02

Lower-cost swap starting card. Sealed cells, 28–40 kg/m³. Mix about 90:100 with PMDI.

04 Polyol technical summary

ISOEXTER TB-307 / KIMpol PE 104 and TEROL 250 / KIMpol PE 103 in the cards, with client TDS numbers. PD 230 stays.

05 Catalysts and supplier memo ranges

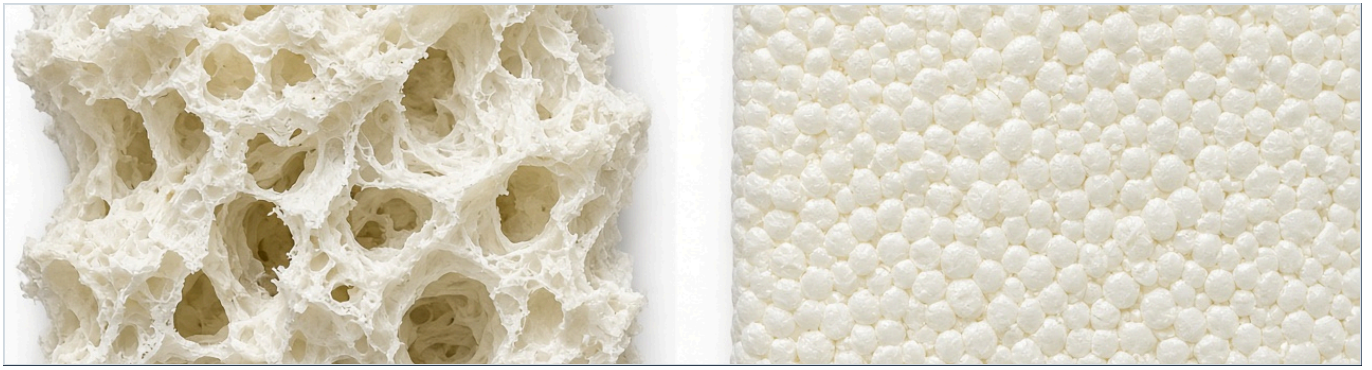
PO 1009433 catalyst roles and xu wang systems A / B / C.

06 Mix method and trial matrix

Cup procedure, index maths, and the first pours — BX-01 first. 50 g weigh sheet is a separate page.

07 Patent notes and sign-off

Honeywell / Arkema numbers from the client screenshots, plus use limits.



Left: open cell — windows broken, air passes, foam stays soft. Right: closed cell — membranes intact, gas trapped, foam is rigid and insulating.

Best Formula · BX-01

BEST

NEW

This is the card to pour first. **HFO-1233zd(E) is mandatory** on closed cell. Memo System C and cup-test v4 are different recipes — do not blend them into this pour.

Why others do not have this

Typical spray copies use Mannich or sucrose for the network, CREAPOL 326 / 1500, and K-15 / DMCHA. They also use HFO — that part is now required here too.

BX-01 still does the rest differently. ISOEXTER TB-307 (f = 2.5) is the network — no Mannich. TEROL 250 (39.5 % aromatic) carries fire. PO 1009433 cats only. **HFO-1233zd(E) 10.0** is the blowing gas. Water is cut to 1.20.

Target	
Density	32–38 kg/m³
Closed-cell content	> 90 %
Cream / gel / tack-free	3–6 s / 8–15 s / 15–28 s
Isocyanate	PMDI, 31.0 % NCO
Index	125 (calc.)
Weight ratio	≈ 83 A : 100 B

BX-01 · Closed-cell · HFO mandatory · 100.0 pbw B-side

Recommended first cup. Faster cream/gel pack. Confirm the drum label. Weigh HFO last, cold, then cap. Mix 4–5 s — cream is now 3–6 s.

INGREDIENT	WHY IT IS IN THIS CARD — NOT IN THE USUAL COPIES	PBW	OH#	EQ
ISOEXTER TB-307 / KIMpol PE 104	Lower-cost f = 2.5 rigid network. Replaces Mannich and CREAPOL 326.	41.3	305	0.2245
TEROL 250 / KIMpol PE 103	Lower-cost flow + 39.5 % aromatic fire. Replaces CREAPOL 1500 and extra TCPP.	24.8	260	0.1149
CREAPOL PD 230	Small toughness trim only. No lower-cost match.	4.0	225	0.0160
TCPP	Cut vs typical 12–23. Terol aromatic does part of the fire job.	6.0	0	0
TEP	Viscosity cut for thick PE 104 / TB-307.	4.0	0	0
Reactive FR	Bound phosphorus. Re-index if the grade has OH.	2.0	0*	0*
Emulsifier	Keeps 1.20 water in the thick polyester.	1.75	0	0

INGREDIENT	WHY IT IS IN THIS CARD — NOT IN THE USUAL COPIES	PBW	OH#	EQ
Closed-cell silicone	Sealed cells. Do not use an open-cell silicone.	1.5	0	0
Water	Cut because HFO does the blow. Do not go back to 1.75.	1.20	6233	0.1333
HFO-1233zd(E)	Mandatory closed-cell blowing gas. 0 OH. Weigh last, cold, cap.	10.0	0	0
PC-15	Raised for faster gel. Do not drop back to 1.15.	1.45	—	—
DABCO 2040	Raised for faster gel / cure.	1.10	—	—
CAT-T	Raised for faster cream / blow.	0.55	—	—
PC-37	Raised for faster cream / blow.	0.35	—	—
B-side total		100.00		0.4888

Index at 125

OH eq = 0.4888 / 100 g B (HFO adds no OH)

NCO eq = 0.4888 × 1.25 = 0.6110

PMDI EW 135.5 → **82.8 g A / 100 g B**

Mix at **83 : 100** by weight. If the drum is TEROL 250 (EW 224.4), A-side is about 82.1 g.

50 g lab cup → 50.00 g B + 41.40 g A

HFO handling

HFO-1233zd(E) boils at about 19 °C. Chill the B cup to **15–18 °C**. Weigh HFO last. Cap at once. Hood on. Mix **4–5 s** so you are out before cream. If cream is still > 6 s: +0.15 CAT-T. If gel is still > 15 s: +0.15 PC-15. If it scorches or snaps too fast: –0.10 of each.

BX-02 · Open-cell best BEST NEW

Only if sucrose is on the floor. Commercial density, not the memo's 24-water screen. Still no polyether 1 / 2 and no 23 pbw TCPP.

INGREDIENT	WHY IT IS DIFFERENT	PBW
Sucrose / sucrose polyol	Needed to open and still stand. Memo uses 33; cup-test splits this with two polyethers.	38.0
TEROL 250 / KIMpol PE 103	Lower-cost aromatic fire. Others use CREAPOL 1500 or none.	16.0
TEP	Viscosity + phosphorus. Cut slightly to make room for faster cats.	20.9
Reactive FR	Bound FR.	7.0

INGREDIENT	WHY IT IS DIFFERENT	PBW
Ecosurf SA-9	Emulsifies 5.2 water. Lower than memo 6.0.	4.0
Open-cell silicone	Opens windows.	1.5
Water	5.2 — between memo 24 and cup-test 4.5. Target 16–22 kg/m ³ .	5.2
PC-37	Raised for faster cream.	1.70
CAT-T	Raised for faster cream.	1.60
PC-15	Raised for faster gel. Watch for closed windows.	2.80
BDMAEE or DEDEE	Raised for faster window rupture.	0.90
Biocide	In-can protection.	0.40
B-side total		100.00

Faster cream/gel pack: target cream 2–4 s. Mix 4–5 s. If cells stay closed: +0.4 water or –0.2 PC-15. If it collapses: +0.3 PC-15. Do not pour BX-02 and OC-01 as one blend. [50 g open-cell weigh sheet → 50.00 g B + 65.70 g A](#)

Open-Cell Foam · OC-01



Open-cell structure. High water makes CO₂. Catalysts blow faster than they gel so the windows rupture.

Target		
Density		7–12 kg/m ³
Closed-cell content		< 10 %
Cream / rise / tack-free		2–5 s / 8–15 s / 10–20 s
Isocyanate		PMDI, 31.0 % NCO
Index, lab cup		90–105
Index, 1:1 spray		35–55

OC-01 · 100.0 pbw B-side

Midpoints of supplier System C. Amine package is 10.0 pbw: blow (PC-37 + CAT-T + BDMAEE) over gel (PC-15).

INGREDIENT	WHY IT IS IN THE FORMULA	PBW
Sucrose / sucrose polyol	High-functionality backbone. Builds a network that can open and still stand.	33.0
Reactive flame retardant	FR that binds into the polymer instead of only sitting in the blend.	8.0
TEP	Cuts viscosity and adds phosphorus FR.	17.0
Ecosurf SA-9	Emulsifies water into the polyol so blow is even.	6.0
Silicone surfactant	Opens and evens the cells. Too little = collapse. Too much = closed windows.	1.2
Water	Main blowing agent. Reacts with NCO to make CO ₂ . This is what makes the foam open and light.	24.0
PC-37	Blow catalyst.	2.4

INGREDIENT	WHY IT IS IN THE FORMULA	PBW
CAT-T	Blow catalyst.	2.4
PC-15	Gel / balance. Holds the rise so it does not collapse.	4.2
BDMAEE or DEDEE	Strong open-cell blow. Helps rupture windows.	1.0
Biocide	In-can protection of the high-water blend.	0.8
B-side total		100.0

OC-02 · If sucrose / polyether is not available

INGREDIENT	PBW	INGREDIENT	PBW
CREAPOL PD 230	40.0	TEROL 250 / KIMpol PE 103	18.0
Emulsifier NP-9 / SA-9	8.0	Silicone surfactant	1.5
Water	20.0	PC-37 + CAT-T	2.0 + 1.8
PC-15	3.5	DABCO 2040 optional	0.3

OC-02 is a 200 g cup trial only. Polyester-only open cell is tighter and more brittle than OC-01. If cells stay closed: +1.0 water or -0.3 PC-15. Do not scale OC-02.

Closed-Cell Foam · CC-01



Closed-cell structure. Low water, higher index, and gel-heavy catalyst keep the membranes intact.

Target	
Density	28–40 kg/m ³
Closed-cell content	> 90 %
Cream / gel / tack-free	8–15 s / 25–45 s / 40–70 s
Isocyanate	PMDI, 31.0 % NCO
Index	115–125 (calc. 120)
Weight ratio	≈ 90 A : 100 B

CC-01 · 100.0 pbw B-side

Water is 2.0 pbw, not 15–26. That is the difference between a closed-cell rigid foam and an open-cell spray foam.

INGREDIENT	WHY IT IS IN THE FORMULA	PBW	OH#	EQ
ISOEXTER TB-307 / KIMpol PE 104	Lower-cost high-OH rigid backbone. Replaces CREAPOL 326 GP.	35.5	305	0.1930
TEROL 250 / KIMpol PE 103	Lower-cost flow polyester. Replaces CREAPOL 1500. Carries pentane in CC-02.	24.0	260	0.1112
CREAPOL PD 230	Softer glycol polyester. Stops the blend going too brittle.	7.0	225	0.0281
TCP	Halogen FR.	12.0	0	0
TEP or DEHMP	FR plus viscosity cut.	6.5	0	0
Reactive FR	Bound FR. Re-index if it has OH groups.	5.0	0*	0*
Emulsifier	Keeps water and FR mixed.	3.5	0	0

INGREDIENT	WHY IT IS IN THE FORMULA	PBW	OH#	EQ
Silicone surfactant	Fine, sealed cells. Do not use an open-cell silicone here.	1.4	0	0
Water	Limited CO ₂ . Enough to help nucleation, not enough to open cells.	2.0	6233	0.2222
PC-15	Gel / balance.	1.30	—	—
DABCO 2040	Strong gel / cure.	1.00	—	—
CAT-T	Small blow trim only.	0.25	—	—
PC-37	Small blow trim only.	0.15	—	—
B-side total		100.0		0.5545

Index at 120

OH eq = 0.5545 / 100 g B

NCO eq = 0.5545 × 1.20 = 0.6654

PMDI EW 135.5 → **90.2 g A / 100 g B**

Mix at **90 : 100** by weight. Confirm volume after you measure densities.

HFO is mandatory

Do not pour this screen without HFO. Prefer **BX-01** (HFO 10.0 already in the 100). If you still pour CC-01: cut water to **1.4** and add **10.0** HFO-1233zd(E) last, cold, then re-index. No water-only closed-cell cup.

This is the lower-cost swap **screen**, not the best card. Pour **BX-01** first (HFO 10.0 mandatory, faster cats, 83 A : 100 B at index 125). If you pour this screen, add HFO and cut water — no water-only closed-cell cup. PD 230 stays — no lower-cost match. Do not put these into OC-01. Confirm the drum label before you pour.

Polyol Technical Summary

CC-01 and OC-02 use TB-307 / TEROL 250, which cost less than CREAPOL 326 / 1500. Mid-range OH figures are used only for the CC-01 index. Specification windows stay as published.

PROPERTY	TB-307 / PE 104	TEROL 250 / PE 103	CREAPOL PD 230
Replaces	CREAPOL 326 GP	CREAPOL 1500	Kept — no lower-cost match
Manufacturer NEW	COIM ISOEXTER TB-307 · Kimpur KIMpol PE 104	Huntsman TEROL 250 · Kimpur KIMpol PE 103	Flokser CREAPOL
Chemistry	Aromatic polyester, higher-OH rigid	Aromatic polyester for rigid PUR / PIR	Aromatic, glycol-based polyester polyol
Stated use	Spray, systems, bunstock. Fire resistance plus mechanical properties.	Spray, pour-in-place, panels, rigid foam	General aromatic polyester modifier
Hydroxyl number	295–315 mg KOH/g (both)	PE 103: 250–270 TEROL 250: 235–255	210–240 mg KOH/g
Mid OH used in calc.	305 mg KOH/g	260 mg KOH/g (PE 103 mid)	225 mg KOH/g
Equivalent weight	183.9 g/eq @ 305	Card: 215.8 @ 260 TEROL published: 224.4	249.3 g/eq
Viscosity @ 25 °C	TB-307: 7 000 cP PE 104: 7 500–8 500 mPa·s	PE 103: 4 000–5 000 mPa·s TEROL: 4 000–6 000 cP	4250–4750 cP
Functionality NEW	TB-307: 2.5 · PE 104: not published	TEROL 250: 2.0 · PE 103: high-functionality	—
Density / SG NEW	TB-307: 1.20 g/cm³	TEROL 250: 1.234 @ 25 °C	—
Acid number NEW	TB-307: max 1.50 mg KOH/g	—	≤ 1 mg KOH/g
Aromatic content NEW	—	TEROL 250: 39.5 %	—
Storage / shelf NEW	Request official TDS	TEROL: 15–50 °C · 6 months	20–40 °C · 12 months

PROPERTY	TB-307 / PE 104	TEROL 250 / PE 103	CREAPOL PD 230
Role in the formulas	CC-01 rigid backbone	CC-01 flow / pentane carrier · OC-02	CC-01 modifier · OC-02 trial

Client TDS pack — four grades, not two drums NEW

Copied from the Sep 2026 technical-data summaries. TB-307 is not PE 104, and TEROL 250 is not PE 103. They are same-role, lower-cost grades. These PDFs are source-derived summaries, not manufacturer-issued original TDS.

PROPERTY	ISOEXTER TB-307	KIMPOL PE 104	TEROL 250	KIMPOL PE 103
Maker	COIM Group	Kimpur	Huntsman	Kimpur
OH#	295–315	295–315	235–255	250–270
Viscosity	7 000 cP	7 500–8 500 mPa·s	4 000–6 000 cP @ 25 °C	4 000–5 000 mPa·s
Functionality	2.5	—	2.0	High-functionality
Density / SG	1.20 g/cm ³	—	1.234 @ 25 °C	—
Other	Acid max 1.50 · spray / systems / bunstock	High fire resistance + mechanical	EW 224.4 · aromatic 39.5 % · 15–50 °C · 6 months	Fire-resistant general-purpose rigid



Keep drums sealed. TEROL 250: 15–50 °C, 6 months. Measure moisture before every trial series.

Equivalent-weight formula

EW (g/eq) = 56 100 ÷ OH number (mg KOH/g)

Water EW = 9.0 g/eq (OH# 6 233)

PMDI EW @ 31.0 % NCO = 135.5 g/eq

How this feeds the formulas

BX-01 is the best closed-cell card: TB-307 (f = 2.5 network), TEROL 250 (aromatic fire), **HFO-1233zd(E) 10.0 mandatory**, faster cat pack (cream 3–6 s / gel 8–15 s), water 1.20,

index 125 → 83 A : 100 B. CC-01 is only a screen and must also take HFO.

NEW If the drum is TEROL 250, use published EW 224.4 (OH 250) — A-side falls to about **89.5 g / 100 g B** at index 120. If the drum is PE 103, keep the card at OH 260. PE 103 is high-functionality; TEROL 250 is $f = 2.0$, so PE 103 may gel faster.

PD 230 remains Flokser Kimya, Hadımköy Mah. Ürgüplü Cad. No: 7/3, 34555 Arnavutköy — İstanbul, Turkey. Request the current official TDS from COIM, Huntsman or Kimpur before scale-up. Summary PDFs are not a performance guarantee.

Replaced Flokser grades — not in the cards

Kept as TDS reference only. Same role, higher cost. Use only if TB-307 / TEROL 250 is not on the floor; then restore the old 1:1 CC-01 index (OH eq 0.6177 → 100.4 g A / 100 g B).

FLOKSER GRADE	OH#	VISCOSITY @ 25 °C	REPLACED BY	WHY IT CAME OUT
CREAPOL 326 GP	380–430 (mid 405)	3500–4500 cP	ISOEXTER TB-307 / KIMpol PE 104	Closest lower-cost high-OH rigid polyester. Lower OH (305 vs 405) and thicker, so the card was re-indexed.
CREAPOL 1500	250–270 (mid 260)	1800–2200 cP	TEROL 250 / KIMpol PE 103	Same OH window. Costs less than CREAPOL 1500. Both are thicker than 1500 — expect a heavier B-side.

Catalysts and Supplier Ranges

Roles from PO 1009433. Chemical names are the industry spellings (the phone photos contain OCR typos). Quantities are listed only so the lab knows which drums are on the floor.

CODE	CHEMICAL NAME	ROLE IN FORMULA	ON THE PO	PACK	HS
PC-15	N,N,N',N'-Tetramethyldipropylenetriamine	Gel / balance	26 236 lb	1 874 lb × 14 totes	2921
PC-37	2-[2-(Dimethylamino)ethoxy]ethanol	Blow	1 984 lb	1 984 lb × 1 tote	2922.19
CAT-T	Trimethyl hydroxyethyl ethylenediamine	Blow	9 370 lb	1 874 lb × 5 totes	2922.1

Memo rule — most open cell	
Blow	CAT-T / BDMAEE / PC-37 / DEDEE
Gel / balance	PC-15
Strong gel / cure	DABCO 2040
Combined OC cat package	USD 5.15 / kg CIF Turkey port

Still required for the formulas
Polymeric MDI (≈ 31 % NCO), silicone surfactant, emulsifier (NP-9 or Ecosurf SA-9), flame retardant (TCPP / TEP / DEHMP / reactive FR), biocide, and — for OC-01 — sucrose / polyether polyols. For every closed-cell cup, HFO-1233zd(E) is mandatory .

Supplier memo — OC/CC ingredient windows (pbw)

From Memo Style.pdf, 31 August 2026, xu wang to george@bazzfoam.com. These are ranges, not a finished 100-part formula. High water (15–26) marks A / B / C as open-cell type. Closed-cell water is 2.0 in CC-01.

INGREDIENT	A	B	C
Sucrose / sucrose polyol	20–60	—	30–36
Polyether polyol 1	—	15–19	—
Polyether polyol 2	—	14–18	—
Reactive FR	—	—	6–10
DEHMP / organophosphate	—	13–17	—

INGREDIENT	A	B	C
TEP	—	—	15–19
TCPP	15–25	—	—
Amine catalysts	4–10	8–12	8–12
Emulsifier	5–15	8–12 NP-9	4–8 Ecosurf SA-9
Silicone surfactant	0.5–2	0.5–2	0.5–2
Water	15–25	18–21	22–26
Biocide	—	—	0.5–1

Mix Method and Trial Matrix



Part B is the formula above. Part A is polymeric MDI. Bring both to temperature, then mix before cream.

Cup method

- 1. Condition A and B at **21–23 °C**.
- 2. Weigh B, then weigh A, into dry cups.
- 3. Pour A into B. Mix **4–5 s** at 2 500–3 000 rpm for both OC and CC — cream is now 2–6 s.
- 4. Clock cream, gel, rise and tack-free from mix start.
- 5. Do not touch after cream. Cut at 24 h.

First cup: 50.00 g B (includes 5.00 g HFO) + 41.40 g A on the [50 g lab](#) sheet. No machine spray until three clean cups.

Working equations

$$EW_{polyol} = 56\,100 / OH\# \cdot eq = pbw / EW \cdot EW_{water} = 9.0 \cdot EW_{PMDI} = 4\,200 / \%NCO$$
$$Index = (NCO\,eq / OH\,eq) \times 100 \cdot A\text{-side (g)} = OH\,eq \times (Index / 100) \times EW_{PMDI}$$

First pours — BX-01 first, then screens

TRIAL	CHANGE FROM CONTROL	WHAT YOU ARE ASKING
BX-01	Best card — HFO 10.0 · faster cat pack	Cream 3–6 s · gel 8–15 s · does it seal?
BX-01a	HFO 10.0 → 11.0 / water 1.20 → 1.10	Lighter / more blow
BX-01b	CAT-T +0.15 / PC-15 +0.15	Still faster if 3–6 / 8–15 is missed
BX-02	Best OC, only if sucrose is on the floor	Commercial 16–22 kg/m³ open cell without polyethers
OC-01	Memo screen	Does System C open at 7–12 kg/m³?
CC-01	Lower-cost swap 90:100 screen	Fallback if BX-01 is too tight

Record

Temperature, mix time, cream, gel, rise, tack-free, collapse, shrink, scorch, friability.

Measure

Free-rise density. Then compressive strength, 70 °C / 90 % RH stability, closed-cell content if you can.

Pass

Three cups, no collapse, density in window, cream/rise pair repeats.

These are laboratory starting formulas, not a production specification or a fire certificate. Confirm silicone grade (open-cell vs closed-cell), FR grade, and the exact PMDI %NCO before you lock a plant card.

Patent Notes and Sign-off

Numbers read from the client screenshots. Listed as blowing-agent / foam technology context, not freedom-to-operate advice.

APPLICANT (AS SHOWN)	PUBLICATION / PATENT	DATE ON SECOND SCREENSHOT
Honeywell	US7655610 / US7655610B2	2024.10
Honeywell	US8618040 / US8618040B2	2023.11
Honeywell	US8962707 / US8962707B2	2023.10
Honeywell	US9884946	2025.4.6
Honeywell	US8053404B2	2024.4
Honeywell	US8623808B2	2024.4
Arkema	CN105001440A	—
Honeywell	EP2640800A2 · EP2737006A2 · EP2606086A2 · EP3567072A1	—
Honeywell	JP5504161B2 · JP6231496B2	—

Source files used

ISOEXTER TB-307, KIMpol PE 103 / PE 104, TEROL 250 TDS summaries · TB307_PE104_TEROL250_PE103_TDS_Summary_Pack.pdf · CREAPOL 326 / 1500 / PD 230 TDS (reference) · Memo Style.pdf (xu wang, 31 Aug 2026) · PO 1009433 catalyst photos · Honeywell / Arkema patent screenshots.

What this pack is not

Not a production specification, fire-test certificate, spray-machine card, or legal clearance. Write the exact silicone, FR and PMDI grades on the trial sheet when the drums are opened.

Next actions

STEP	ACTION	OWNER
1	Confirm sucrose / polyether, silicone, FR, PMDI, and which floor drum is open (TB-307 vs PE 104, TEROL 250 vs PE 103).	Purchasing
2	Run BX-01 first as a 200–300 g cup at 22 °C. Then OC-01 / CC-01 only as screens.	Lab

STEP	ACTION	OWNER
3	Change only one variable per trial.	Lab
4	Closed-cell cups must include HFO-1233zd(E). Hood on. Chill B. Weigh HFO last.	Lab / HSE
5	Return cream / rise / density numbers for the next revision.	Technical

Issue and sign-off

ROLE	NAME	SIGNATURE	DATE
Prepared	Technical — Bazz Foam formulation pack Rev. J		28 Sep 2026
Issued to	Bazzfoam		
Lab acceptance	First cup series BX-01 complete		
Next revision	After cream / rise / density return		

Prepared from the supplied TDS files, PO photographs and the xu wang memo. Flokser, COIM, ISOEXTER, Huntsman, TEROL, Kimpur, KIMpol, Honeywell, Arkema, DABCO and CREAPOL names are the property of their owners. Commercial use needs controlled lab validation.